Diffusion Processes for Inference of Structural Causal Models

IA pour la Science

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Inference of Structural Causal Models

The goal is to infer an SCM-model f_G with optimal

1) complexity $\mathcal{L}_{\mathsf{C}}$,
2) accuracy $\mathcal{L}_{\mathbf{D}}$,
3) robustness $\mathcal{L}_{\mathsf{ci}}$.

The models:

1) the model f_{ai} maps a data set $\mathbf{D}$ to a model $f_G(\mathbf{w}, \mathbf{x})$,
2) the model f_G maps a sample $\mathbf{x} \in \mathbf{D}$ to its forecast $\hat{\mathbf{x}}$, provided optimal parameters $\hat{\mathbf{w}}$.

The van der Pol oscillator, the mathematical model

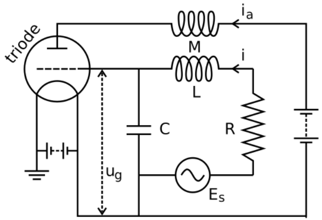
The forced Van der Pol oscillator:

$$\frac{d^2x}{dt^2} - \mu(1-x^2)\frac{dx}{dt} + x - A\sin(\omega t) = 0,$$

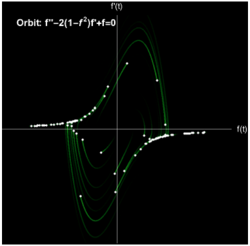
has two linear and one non-linear parameters: μ, A, ω, where A is the amplitude, of the wave function and ω is its angular velocity. The causal graph has four vertices, the variables:

$$X_1, X_2, X_3 \mapsto \ddot{x}, \dot{x}, x$$

and the time t.

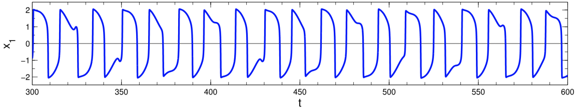


Electrical circuit resulting in a forced Van der Pol oscillator

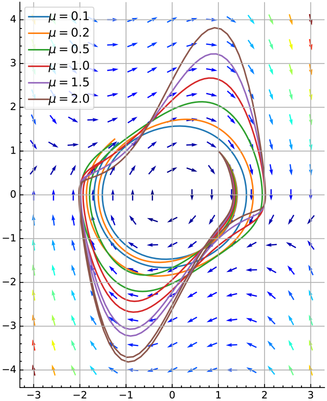


Randomly chosen initial conditions are attracted to a stable orbit

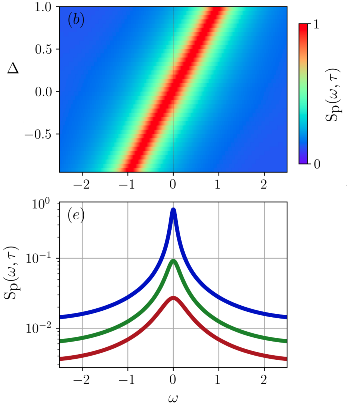
Three spaces to observe the Van der Pol oscillator



Chaotic behaviour in the Van der Pol oscillator with sinusoidal forcing.



Phase portrait of the Van der Pol's equation



Oscillator power spectra with two parameters

Alternative form of the Van der Pol oscillator model

An alternative form is a time-independent Hamiltonian by augmenting it to a four-dimensional, $X_1, \ldots, X_4$ autonomous dynamical system using an auxiliary second-order nonlinear differential equation:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \qquad \ddot{y} + \mu(1 - x^2)\dot{y} + y = 0.$$

The Hamiltonian

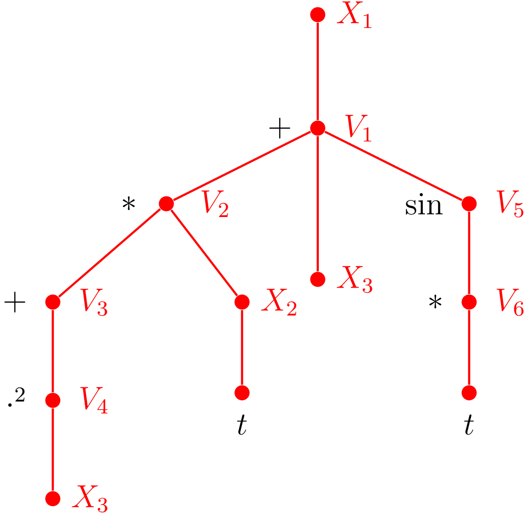
$$H(x, y, p_x, p_y) = p_x p_y + xy - \mu(1 - x^2) y p_y,$$

where

$$p_x = \dot{y} + \mu(1 - x^2)y \quad \text{and} \quad p_y = \dot{x}$$

are the conjugate momenta corresponding to «x» and «y».

The causality graph of the Van der Pol oscillator model



$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x - A\sin(\omega t) = 0,$$

$$X_1(t) = \mu(1 - X_3^2) * X_2(t) - X_3 + A\sin(\omega * t).$$

The adjacent matrix Z

$$X_1(t) = \mu(1 - X_3^2) \times X_2(t) - X_3 + A\sin(\omega t).$$

	X_1	X_2	X_3	V_1	V_2	V_3	V_4	V_5	V_6	t
X_1				1						1
X_2										1
X_3										
V_1			1		1			1		
V_2		1				1				
V_3							1			
V_4			1							
V_5									1	
V_6										1

The basic functions for the Van der Pol model

The nodes: $X_1(t), X_2(t), X_3$ are the observable variables.

The latent variables $V_1, \dots, V_6$ are in the set (the dictionary)

$$\{\sin(\cdot), (\cdot)^2, +(w_1, \dots, w_{k+1}), \times(\dots, w_0)\}.$$

The function «$+$» has $k+1$ linear parameters and k arguments.
The function «$\times$» has one parameters and k' arguments.

The causal model for the graph G

$f_G(\mathbf{w}) : \mathbf{x}_t \mapsto \hat{\mathbf{x}}_t$ with observable variables $(X_2, X_3)(t) \mapsto \hat{X}_1(t)$

with parameters $\mathbf{w}$ and data $\mathbf{x}$ has the form

$$X_1(t) = w_1(w_4 - w_5 X_3^2) \times X_2(t) - w_2 X_3 + w_3 \sin(w_6 t).$$

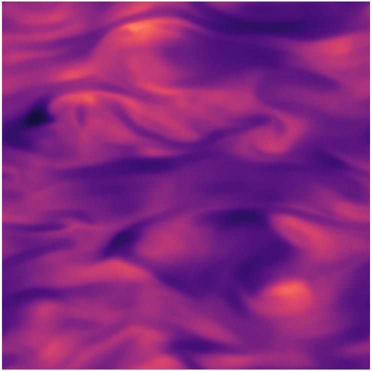
Magnetohydrodynamics compressible turbulence[1]

MHD fluid flows in the compressible limit

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0,$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v}\mathbf{v} - \mathbf{B}\mathbf{B}) + \nabla p = 0,$$

$$\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) = 0,$$



where ρ is the density, $\mathbf{v}$ is the velocity, $\mathbf{B}$ is the magnetic field (note that the $\nabla \cdot (\rho \mathbf{v}\mathbf{v} - \mathbf{B}\mathbf{B}) \in \mathbb{R}^3$), and p is the gas pressure.

[1] The Well, MHD dataset, 2025

The MHD data and CI variables

Six variables $X = X(t, s)$ with cod in $\mathbb{R}^1$ or $\mathbb{R}^3$

$$X_1, X_2, X_3 = \rho, \mathbf{v}, \mathbf{B}, \qquad X_4, X_5, X_6 = \partial\rho, \partial(\rho\mathbf{v}), \partial\mathbf{B}.$$

Dictionary

comprises the functions $+, *, \times$ and the operators

$$\nabla \cdot \mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^1 \quad \text{and} \quad \nabla \times \mathbf{F} : \mathbb{R}^1 \to \mathbb{R}^3.$$

Data 4.58TB

Time $t \in [1, \ldots, 100]$, space $s \in [1, \ldots, 256]^3$.

Density $\rho(t, s) \in \mathbb{R}^1_+$, velocity $\mathbf{v}(t, s) \in \mathbb{R}^3$, magnetic field $\mathbf{B}(t, s) \in \mathbb{R}^3$ are scalar and vector fields.

Simulations: 10 Initial conditions $\times$ 10 combination of parameters make 100 trajectories.

The graph generation and rewriting

A graph $G(V, E)$ with the

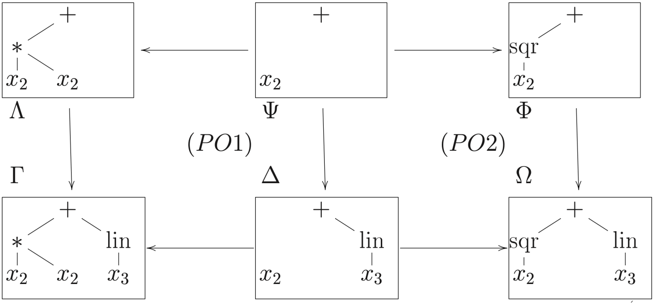
1) coloring function $c(V)$, where c is in the dictionary,
2) weights $\mathbf{w}(V)$,
3) probability of the vertices $p(V)$,
4) probability of the edges $\mathbf{Z} \in \{0, 1\}^{n \times n}$ or $\mathbf{\Gamma} \in [0, 1]^{n \times n}$.

The graph G defined the function $f_G(\mathbf{w}, \mathbf{x}) \to \hat{\mathbf{x}}$.

There given a graph rewriting rules $G \mapsto G'$ so that each step delivers an admissible graph (admissible means a graph G defines a function f_G).

To produce a sequence of the graphs introduce a vector $\mathbf{z} = \textbf{vec}(\mathbf{Z})$.

Graph rewriting ensures the admissibility of the model f_G [2]



A graph of the form $+(\times(+(\ldots)))$ could be simplified, too.

$$\begin{tikzcd}
\Lambda_1 \arrow[d, "\mathfrak{n}_1"] & \Psi_1 \arrow[l, "l_1"'] \arrow[r, "r_1"] \arrow[d, "k_1"] & \Phi_1 \arrow[dr, "\mathfrak{n}_1"] \arrow[drrr, dashed, "i"] & & \Lambda_2 \arrow[dl, "\mathfrak{m}_2"'] \arrow[dlll, dashed, "j"'] & \Psi_2 \arrow[l, "l_2"'] \arrow[r, "r_2"] \arrow[d, "k_2"] & \Phi_2 \arrow[d, "\mathfrak{n}_2"] \\
\Gamma & \Delta_1 \arrow[l, "f_1"'] \arrow[rr, "g_1"] & & \Omega & & \Delta_2 \arrow[ll, "f_2"'] \arrow[r, "g_2"] & \Gamma'
\end{tikzcd}$$

[2] This topic will be investigated separately. See Hartmut Ehrig et al. Fundamentals of Algebraic Graph Transformation, Springer, 2006.

Coherent the SCM inference (problem statement)

The given data and the problem

1. There given a collection $\{\mathbf{D}\} \rightarrowtail \{f_G\}$ with set injection.
2. For any given data set $\mathbf{D}$ the model $f_{\mathsf{ai}} : \mathbf{D} \mapsto f_G$ such that
3. The model $f_G(\mathbf{w}) : \mathbf{x} \mapsto \hat{\mathbf{x}}$ with optimal criteria $(\mathcal{L}_{\mathsf{C}}, \mathcal{L}_{\mathbf{D}}, \mathcal{L}_{\mathsf{ci}})$.
4. A context or prior information $p(f_G)$ is provided along with the data $\mathbf{D}$ as a vector.

Coherent the SCM inference (addendum)

Consigns

1. Denote the given f_G as f_0.
2. Convert its given graph $G(V,E)(\mathfrak{c},\mathfrak{r})$ to the vector $\mathbf{z}_0$.
3. The rules $\mathfrak{r}$ rewrite the graph G; the colors $\mathfrak{c}$ apply restrictions to the adjacency matrix, $\mathbf{z} = \textbf{vec}(\mathbf{Z})$, the number of 1s in its row must coincide to the corresponding dictionary function.
4. The DAG G splits the variables $X_1,\dots,X_n$ to the source and target variables. We denote the corresponding data samples by $\mathbf{x} \mapsto \hat{\mathbf{x}}$ without collision.
5. We treat all types of variables, $X, X(t), X(t,s)$ as random processes, each variable has its own dimensionality. For example, the MHD treat most of variables as a process $X(s,t)$ a vector function $\mathbb{R}^4 \to \mathbb{R}^3$.

Coherent the SCM inference (algorithm)

The algorithm with steps $t = [T, \dots, 0]$

1. Sample some new structure $\mathbf{z}_t$ from the distribution $q(\mathbf{z}_t \mid \mathbf{z}_{t-1})q(\mathbf{z}_{t-1} \mid \mathbf{A}_{t-1})q(f_G)$.
2. Convert the structure and its parameters $(\mathbf{z}, \mathbf{w})$ to the SCM $f_G(\mathbf{w}, \mathbf{x})$.
3. Rewrite the graph G of the SCM according to the rules $\mathfrak{r}$.
4. Note that the initial values of parameters $\mathbf{w}_t$ of f_t are inherited from $\hat{\mathbf{w}}_{t-1}$.
5. Estimate the parameters $\hat{\mathbf{w}}$ according to the data $\{\mathbf{x}\}$.
6. Given $\hat{\mathbf{w}}$ estimate the covariance matrix $\hat{\mathbf{A}}$.
7. Make do-intervention and estimate the covariance matrix $\hat{\mathbf{A}}'$.
8. Analyze the couple $\mathbf{A}, \mathbf{A}'$ and set the parameters $\boldsymbol{\pi}$.
9. Repeat until $t = T$. Keep the spectrum of $\mathbf{D}$ at $t = T$.

(continued)

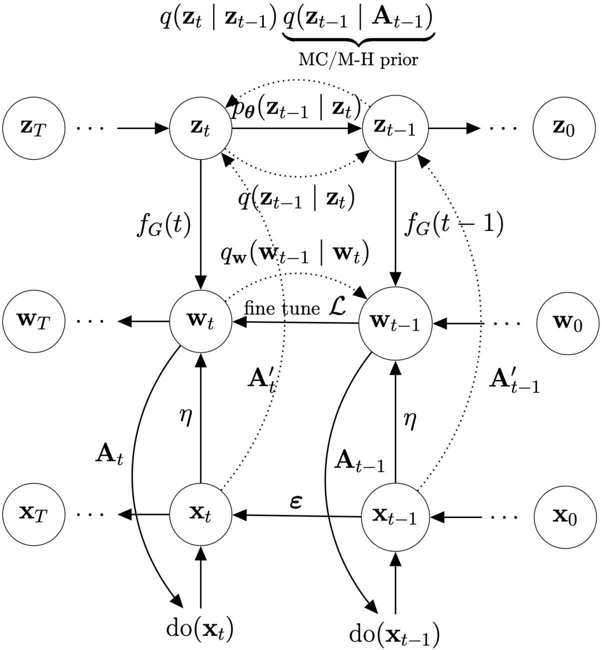
Forecasting

1. Given the data set $\mathbf{D}$ (and its spectrun at $t = T$) and the prior $p(G)$.
2. Make the inference $p_{\boldsymbol{\theta}}(\mathbf{z}_{t-1} \mid \mathbf{z}_t)$ and compute $\mathbf{z}_0$.
3. At each step t reconstruct $\mathbf{w}_t$. No intervention needed.
4. Return $f_G(\hat{\mathbf{w}}, \mathbf{x})$. Return optimal $\mathcal{L}_{\mathbf{D}}$, complexity, and robustness.

Notes

1. Step 4: new parameters are set to deliver it to the dictionary functions. Examples. For $+$ the parameters $\mathbf{w} = [0, 0]$ delivers id : $0x + 0 = x$. For $+$ the parameters $\mathbf{w} = [1]$ delivers id : $1 * x = x$. The cross-product $\times$ delivers $\mathbf{I}$ or the identity tensor is written as the Kronecker delta. There is no parameters that delivers id for functions like $\sin(x), \ln(x)$.
2. Steps 2,3,5,6,7 do not depend on time.

Coherent generation of SCMs with do-intervention



The Metropolis Hastings graph sampling

Sample the new structure $\mathbf{z}_t$ from $q(\mathbf{z}_t \mid \mathbf{z}_{t-1})q(\mathbf{z}_{t-1} \mid \mathbf{A}_{t-1})q(f_G)$.

To sample graphs $\mathbf{z}_0 = \{\mathbf{z} \mid \mathbf{z} \sim p(\mathbf{z}|\boldsymbol{\pi})\}$. The element $\mathbf{z}_{t+1}$ of the sample depends on the previous element $\mathbf{z}_t$ of the sample only.
Denote by $Q(\mathbf{z} \mid \mathbf{z}')$ an auxiliary distribution $Q(\mathbf{z} \mid \mathbf{z}')$.
Choose an initial element $\mathbf{z}_0$ and assign $\mathbf{z}_0 = \{\mathbf{z}_0\}$.
Then let an element $\mathbf{z}_t$ be chosen according to the distribution $Q(\mathbf{z}' \mid \mathbf{a}_t)$. The next element $\mathbf{z}'$ is generated randomly. Then the algorithm computes the acceptance ratio a:

$$a = \min_{\mathbf{z}' \in \mathbb{R}^n} \left(\frac{p(D \mid \mathbf{a}', B_0) \quad Q(\mathbf{z}_t \mid \mathbf{a}')}{p(D \mid \mathbf{a}_t, B_0) \quad Q(\mathbf{z}' \mid \mathbf{z}_t)}, 1 \right).$$

The algorithm accepts the candidate $\mathbf{z}'$ with the probability a,

$$\mathbf{z}_{t+1} = \mathbf{z}', \qquad , W_0 := W_0 \cup \mathbf{z}'.$$

Otherwise, the algorithm rejects the candidate and puts $\mathbf{z}_{t+1} = \mathbf{z}_t$,

$$\mathbf{z}_{t+1} = \begin{cases} \mathbf{z}', & \text{with the probability} \quad a, \\ \mathbf{z}_t, & \text{with the probability}(1-a). \end{cases}$$

The Metropolis Hastings graph sampling

Let the auxiliary distribution $Q(\mathbf{z} \mid \mathbf{z}')$ be normal (it is pretty simplified case; replace to uniform, Dirichlet or Gumbel-Softmax distribution):

$$Q(\mathbf{z} \mid \mathbf{z}') = Q(\mathbf{z}' \mid \mathbf{z}) = \frac{1}{(2\pi\alpha^{-1})^{n/2}} \exp\left(-\frac{\alpha}{2}(\mathbf{z} - \mathbf{z}')^{\mathsf{T}}(\mathbf{z} - \mathbf{z}')\right).$$

That is, the function $Q(\mathbf{z} \mid \mathbf{z}')$ is symmetric and the acceptance ratio a simplifies to

$$a = \min_{\mathbf{z}' \in \mathbb{R}^n} \left(\frac{p(D \mid \mathbf{a}', B_0)}{p(D \mid \mathbf{a}_t, B_0)}, 1\right).$$

The initial element $\mathbf{z}_0$ is chosen randomly from the distribution $p(\mid \boldsymbol{\pi})$.

The do-inference testing procedure

There given

1) the DAG G corresponds to the model f_G,
2) its parameters and covariance matrix $(\hat{\mathbf{w}}, \hat{A})$,
3) the strategy of do-intervention in $\{X\}$ of n steps[3]

Collect the covariance matrices

1. Sample a subset of $\mathbf{x}$ from the dats set $\mathbf{D}$.
2. Set do$X$ on the step $1, \ldots, n$.
3. Estimate the new covariance matrices $\hat{\mathbf{A}}_1, \ldots, \hat{\mathbf{A}}_n$ for each step.
4. Estimate the new aggregate covariance matrix $\hat{\mathbf{A}}$ for each step.
5. Set the parameters for the graph generation $\boldsymbol{\pi}$.

[3] For example, set each input X (of n inputs, according to G) to the constant do$X = \mathbb{E}[X]$.

Discrete genetic algorithm for model selection

1. There is a set of binary vectors $\{\mathbf{a}_1, \dots, \mathbf{a}_P\}, \quad \mathbf{a} \in \{0,1\}^n$.
2. Select two vectors $\mathbf{a}_p,\ \mathbf{a}_q, \quad p, q \in \{1, \dots, P\}$.
3. Choose a random number $\nu \in \{1, \dots, n-1\}$.
4. Split both vectors and exchange their tails:

$$[a_{p,1}, \dots, a_{p,\nu}, a_{q,\nu+1}, \dots, a_{q,n}] \to \mathbf{a}'_p,$$

$$[a_{q,1}, \dots, a_{q,\nu}, a_{p,\nu+1}, \dots, a_{p,n}] \to \mathbf{a}'_q.$$

5. Choose random numbers $\eta_1, \dots, \eta_Q \in \{1, \dots, n\}$.
6. Invert positions $\eta_1, \dots, \eta_Q$ of the vectors $\mathbf{a}'_p,\ \mathbf{a}'_q$.
7. Repeat Steps 2–6 $\frac{P}{2}$ times.
8. Evaluate the obtained models.

Repeat the procedure R times; here P, Q, and R are the parameters of the algorithm, and n is the number of features in the corresponding model.

Plan of the algorithm of the SCM inference

1. Requirements to model
2. How to construct a model
3. Algebraic and Statistical assumptions
4. Do-intervention and testing (test-ability)
5. Local and global modeling expand
6. Graph diffusion and reconstruction
7. Neural ODE is possible
8. Fine-tuning
9. Testing and do-inference
10. The general probabilistic model and loss function
11. Non-parametric modeling and Optimal sampling
12. Time and space segmentation

Models

The goal is to learn to search a model of optimal structure.

The structure must be admissible:

1. An acyclic directed graph with given number of roots and leafs.
2. Its nodes are operations from a given set.
3. It must fit the data and respect statistical assumptions about the data.

Two main approaches are:

1. Symbolic regression is another computationally complex genetic algorithm.
2. Structure learning shows difficulties in delivering admissible structures of models.

The slides below lists variants of problem statement and methods to solve it.